

EDUGENIE — SCALABILITY & FUTURE PLAN

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CURRENT LIMITATIONS

- Runs only on localhost
- Text-only input (no voice, image, or PDF)
- Single language (English)
- No user accounts or progress tracking
- No conversation memory (each request is independent)

SCALABILITY ROADMAP

Phase 1 (Next 1 month)

- Deploy backend to Render or Railway (free tier)
- Add voice input using the Web Speech API
- Add image input support (OCR-based question solving)
- Implement caching for repeated questions

Phase 2 (3 months)

- User accounts with saved history
- Progress tracking dashboard (topics studied, quizzes taken)
- Multi-language support (Hindi, Tamil, Spanish, French)
- PDF upload for summarization and quiz generation

Phase 3 (6 months)

- Voice-based interaction (two-way conversation)
- Integration with Google Classroom, Moodle, and Canvas
- Gamified elements (badges, streaks, XP points)
- Mobile app using React Native

Phase 4 (1 year)

- Multi-user study groups
- Teacher dashboard for tracking student progress
- Adaptive learning paths based on performance
- Offline mode with local model fallback

TECHNICAL SCALABILITY

- FastAPI handles 1000+ concurrent requests with Gunicorn workers
- Gemini API scales automatically with paid tier
- CDN caching for static assets
- Redis for caching common questions
- PostgreSQL for user data

MONETIZATION (OPTIONAL)

- Free tier: 10 requests per day
- Pro tier: Rs. 149/month for unlimited access
- Institution tier: Custom pricing for schools and colleges
- Enterprise: API access for LMS integration

PERFORMANCE IMPROVEMENTS PLANNED

- **Streaming responses (show answer as it is generated)**
- **Pre-computed answers for common school topics**
- **Caching layer for FAQ-style questions**
- **Reduce prompt size for faster responses**